PROJECT REPORT
OF
INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)

ON

SMART ATTEND: AI-POWERED CLASSROOM ATTENDANCE SYSTEM

Submitted in partial fulfilment of the requirements
for the award of the degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"Smart Attend: AI-Powered Classroom Attendance System"**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering (Computer Science and Engineering)** submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Dr. Gurpreet Singh**.

The matter presented in this project report has not been submitted by us in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

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Furthermore, we extend our heartfelt thanks to our friends and classmates for their cooperation, constructive feedback, and motivation during the project development process. Lastly, we are deeply grateful to our families for their unwavering support, encouragement, and belief in us, which kept us motivated throughout this journey.

ABSTRACT

Smart Attend is an advanced AI-powered classroom attendance and monitoring system designed to automate attendance marking while ensuring authenticity and reliability. Traditional attendance methods, such as manual roll calls and paper-based records, are time-consuming, inefficient, and prone to human errors. Additionally, these methods allow proxy attendance, which affects the accuracy and integrity of academic records. This project aims to overcome these limitations by developing an intelligent and automated solution using modern technologies.

The proposed system leverages computer vision and deep learning-based facial recognition techniques to identify students in real time through a classroom-installed camera. Once a student is detected and recognized, attendance is marked automatically without any manual intervention. To further enhance reliability, the system incorporates real-time location verification using GPS or network-based tracking to ensure that students are physically present within the classroom premises.

Furthermore, the system integrates an AI-based chatbot that assists users by providing attendance information, answering queries, and improving user interaction. It also generates structured attendance reports for faculty and administrative purposes, enabling efficient monitoring and analysis.

Overall, Smart Attend improves efficiency, reduces manual workload, enhances accuracy, and promotes a smart and transparent educational environment.

1.INTRODUCTION

1.1 Problem Background

Attendance management is a critical aspect of educational institutions as it directly influences student evaluation, discipline, and academic performance tracking. However, traditional attendance methods such as manual roll calls and paper-based registers are still widely used in many institutions. These methods are not only time-consuming but also prone to errors caused by human negligence. Additionally, such systems allow proxy attendance, where one student marks attendance on behalf of another, resulting in inaccurate and unreliable records.

Another major limitation of traditional systems is the absence of real-time monitoring and efficient data management. Attendance records are often maintained manually, making it difficult to analyze data or generate reports effectively. With rapid technological advancements and the growing need for automation, there is an increasing demand for a smart attendance system that ensures accuracy, efficiency, and transparency while minimizing human effort.

1.2 Objective of the Project

The primary objective of the Smart Attend system is to develop an intelligent and automated attendance management solution using artificial intelligence and computer vision techniques. The system aims to replace traditional attendance methods with a more efficient, accurate, and reliable approach.

The key objectives of the project are as follows:

- To automate the attendance marking process using facial recognition technology.
- To ensure the physical presence of students through real-time location verification.
- To eliminate the possibility of proxy attendance.
- To reduce the time and effort required for manual attendance recording.
- To improve the accuracy and reliability of attendance data.
- To generate structured attendance reports for better analysis and decision-making.

- To enhance user interaction by integrating an AI-based chatbot for attendance-related queries.

1.3Proposed Solution / System Overview

Smart Attend is an integrated classroom management system that combines advanced technologies such as computer vision, deep learning, and location tracking to automate attendance management. The system utilizes a camera installed in the classroom to capture real-time video streams of students. These video frames are processed using facial recognition algorithms to detect and identify individual students accurately.

Once a student is successfully recognized, the system automatically records their attendance in the database without any manual intervention. To ensure authenticity, the system incorporates a location verification mechanism using GPS or network-based tracking methods. This dual verification approach ensures that attendance is marked only when the student is physically present within the classroom premises.

In addition, the system includes an AI-powered chatbot that provides users with quick access to attendance information and responds to basic queries. The system also generates structured attendance reports that assist faculty members in monitoring attendance patterns and making informed decisions. Overall, the proposed solution is efficient, reliable, scalable, and well-suited for modern educational environments.

METHODOLOGY

The Smart Attend system follows a structured and systematic methodology to ensure accurate, efficient, and automated attendance management. The workflow is divided into multiple stages, starting from data acquisition to report generation, ensuring minimal human intervention and maximum reliability.

Step 1: Data Acquisition

A camera installed in the classroom captures real-time video streams of students. The continuous video feed is divided into frames, which are further processed for analysis.

Step 2: Face Detection

Each frame is processed using computer vision techniques with OpenCV to detect human faces. This step identifies all visible faces present in the classroom environment.

Step 3: Face Recognition

Detected faces are passed to a deep learning-based facial recognition system. The system compares these faces with a pre-trained dataset of registered students. This dataset is created during the initial setup phase by storing multiple images of each student for improved accuracy.

Step 4: Identity Verification

The system matches detected faces with stored data to identify individual students. This ensures that each detected face corresponds to a valid and registered student.

Step 5: Location Verification

To ensure authenticity, the system verifies the student’s physical presence using GPS or network-based location tracking

Step 6: Attendance Marking

Once identity and location verification are successful, attendance is automatically marked. The system records student details along with date and time.\

Step 7: Data Storage

All attendance records are stored in a structured SQLite database. This enables efficient data management, quick retrieval, and secure storage.

Step 8: Report Generation

The stored data is processed to generate attendance reports, statistics, and trends. These reports help faculty members monitor attendance patterns and make informed decisions.

Step 9: Chatbot Interaction

An AI-based chatbot is integrated to allow users to query attendance data. It provides information such as attendance status, percentage, and absence details through natural language interaction.

Overall, this methodology ensures a seamless, automated, and reliable attendance system. The modular architecture allows easy scalability, maintenance, and future enhancements.

TOOLS AND TECHNOLOGIES

The Smart Attend system is developed using a combination of modern tools and technologies that enable efficient implementation, accurate processing, and reliable performance. The key tools and technologies used in the system are described below:

1. Programming Language:

Python: Python is used as the primary programming language due to its simplicity, readability, and extensive support for artificial intelligence, machine learning, and computer vision libraries.

2. Computer Vision and AI Libraries:

- **OpenCV**: Used for real-time image processing and face detection from video streams.

- **face_recognition**: A deep learning-based library used for accurate facial recognition and identification of students.

3. User Interface:

PyQt5: Used to develop a graphical user interface (GUI) that allows users to interact with the system easily and efficiently.

4. Database Management:

SQLite: A lightweight and efficient database used to store attendance records, student details, and system data in a structured format.

5. Location Tracking Technologies:

GPS / Network-Based Location Services: Used to verify the real-time location of students and ensure their physical presence within the classroom premises.

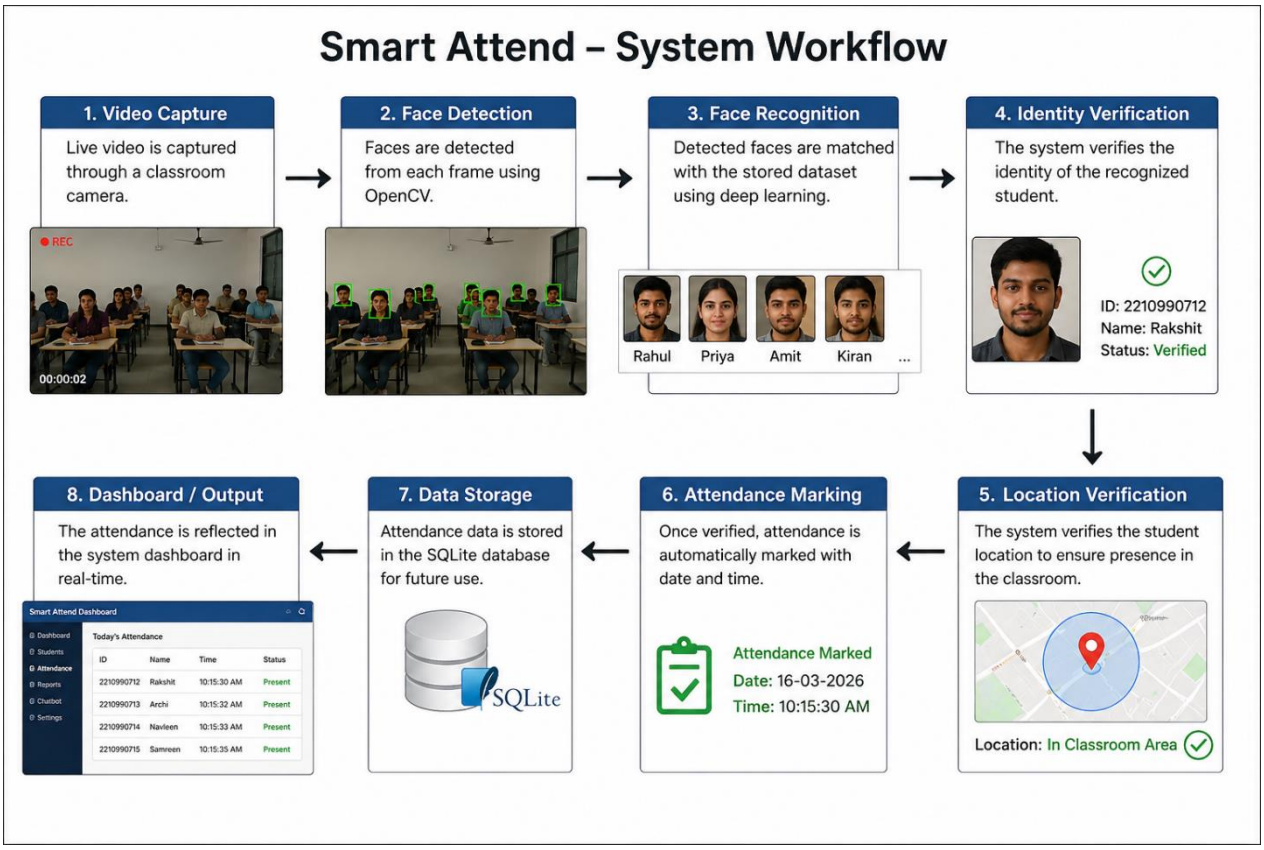
6. Supporting Libraries:**Geocoder / Requests (Optional)**: Used for IP-based location detection and fallback mechanisms.

IMPLEMENTATION

The Smart Attend system is implemented using a modular approach, where different components are developed as independent modules and integrated to provide an efficient and automated attendance management solution. The system is designed to operate in real time, ensuring high accuracy and minimal manual intervention.

1. System Workflow

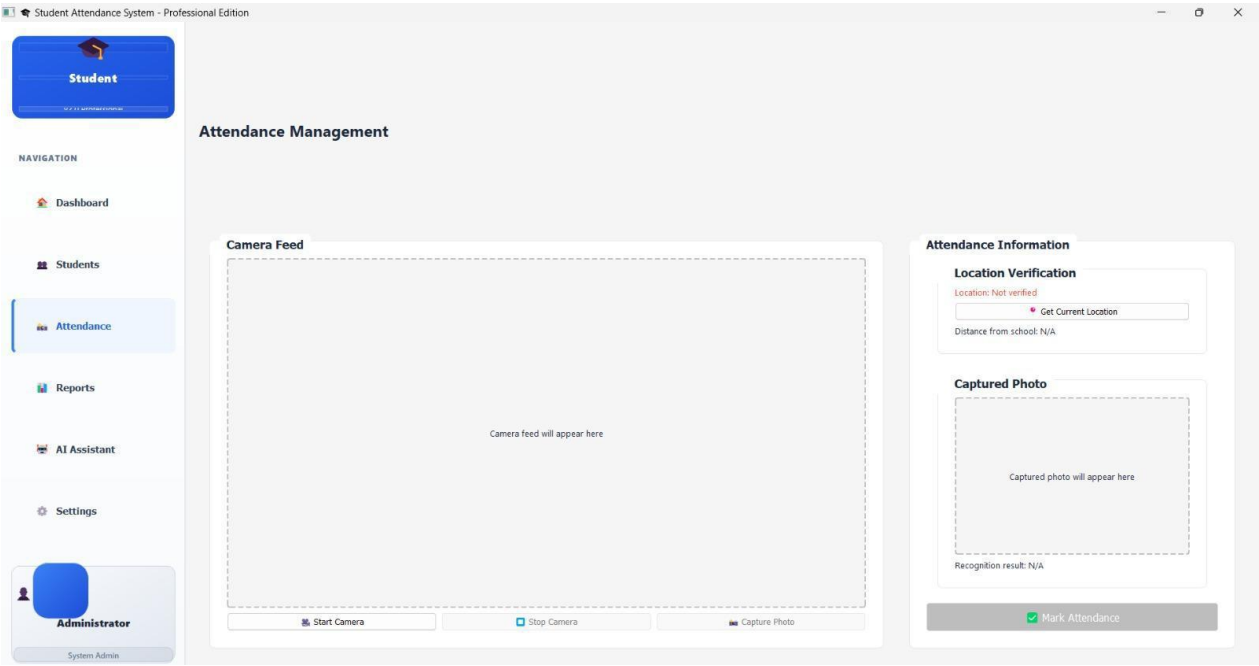
The Smart Attend system follows a real-time workflow starting from video capture to attendance storage. The system captures live video through a camera, detects faces from each frame, matches them with the stored dataset, verifies identity and location, and automatically marks attendance in the database.



2. Face Recognition Module

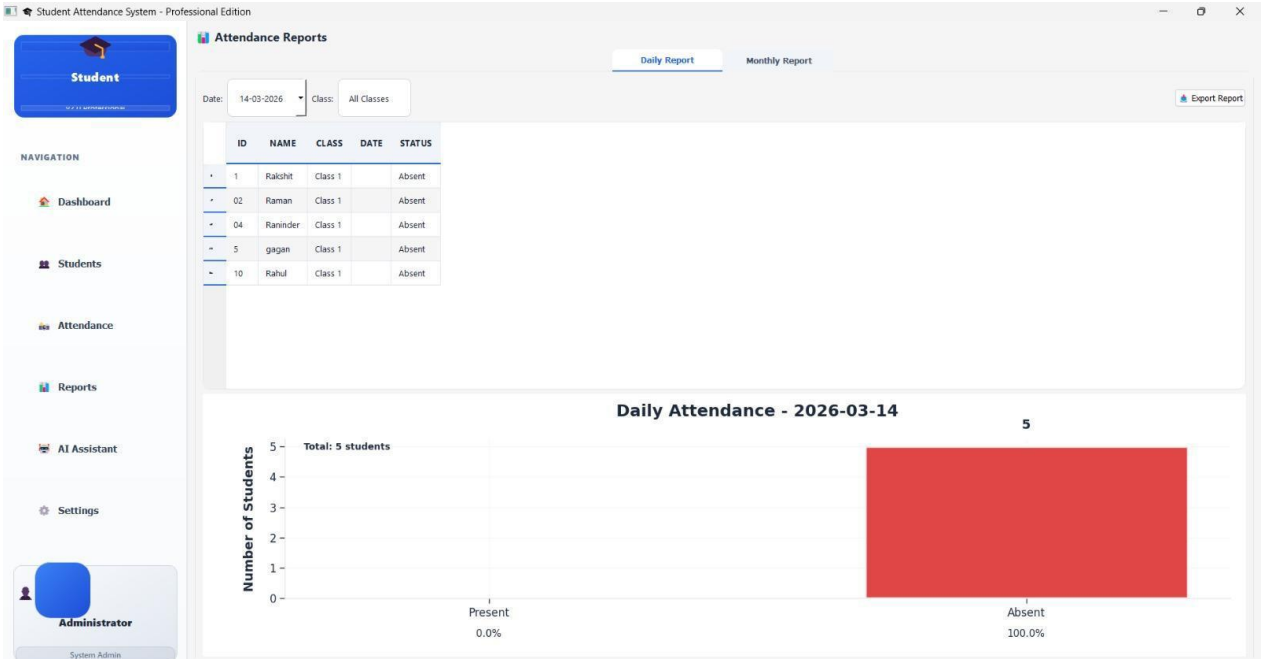
The face recognition module is the core component of the system. It is implemented using OpenCV and the face_recognition library. OpenCV is used to capture live video streams and detect faces in each frame, while the face_recognition library applies deep learning algorithms to identify students.

A dataset of student images is created during the initial setup phase, where multiple images of each student are stored. These images are encoded and used for matching during real-time recognition, ensuring accurate identification under different conditions.



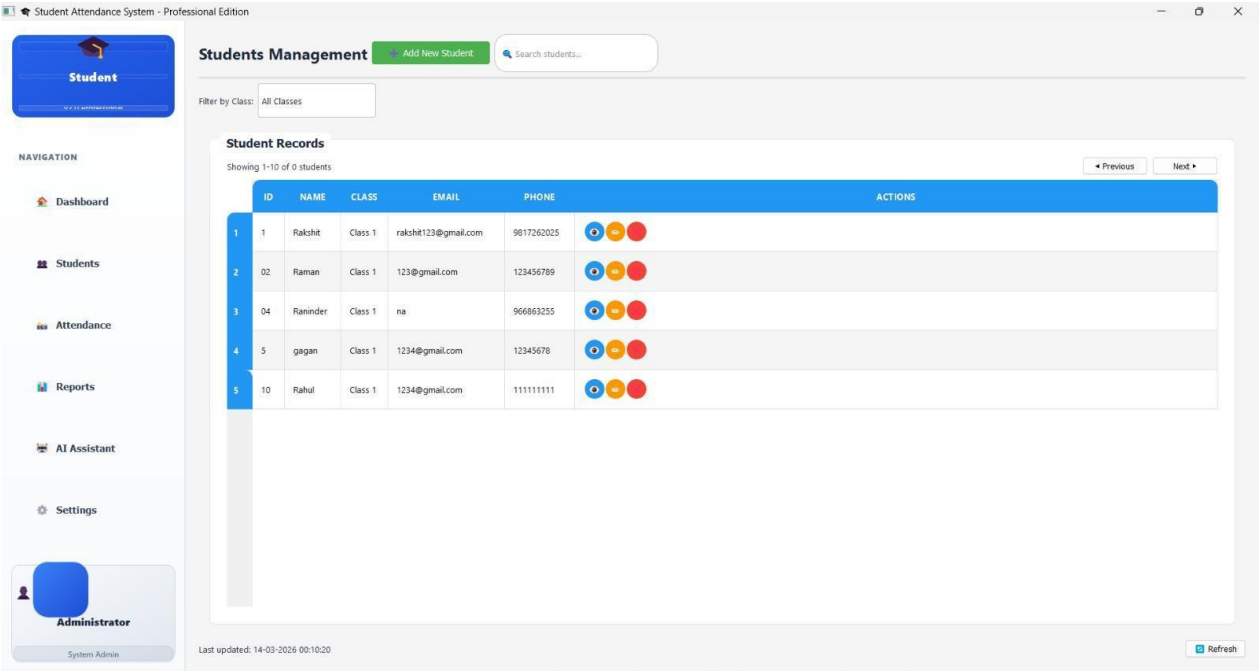
2. Attendance Management Module

Once a student is successfully recognized, the system automatically marks attendance. The attendance includes details such as student ID, name, date, and time. The system also supports updating and editing attendance records when required.



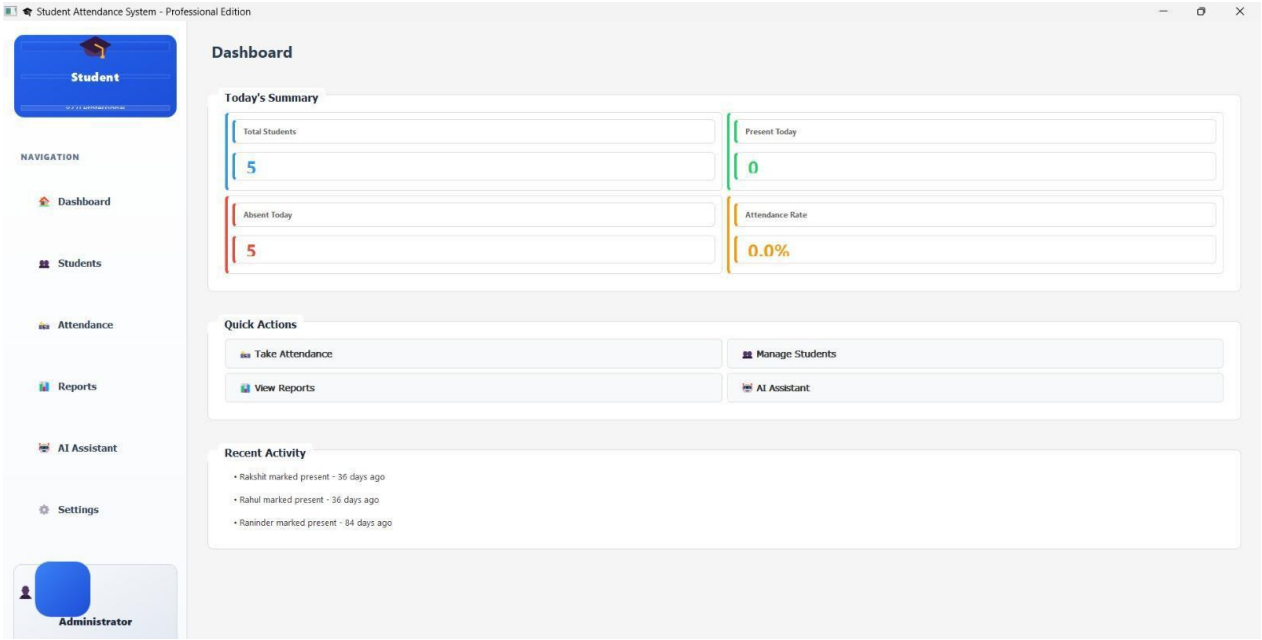
3. Database Module

The system uses SQLite as the backend database for storing student details and attendance records. The database is structured to allow efficient storage, retrieval, and report generation. Tables are designed to manage student information and daily attendance logs.



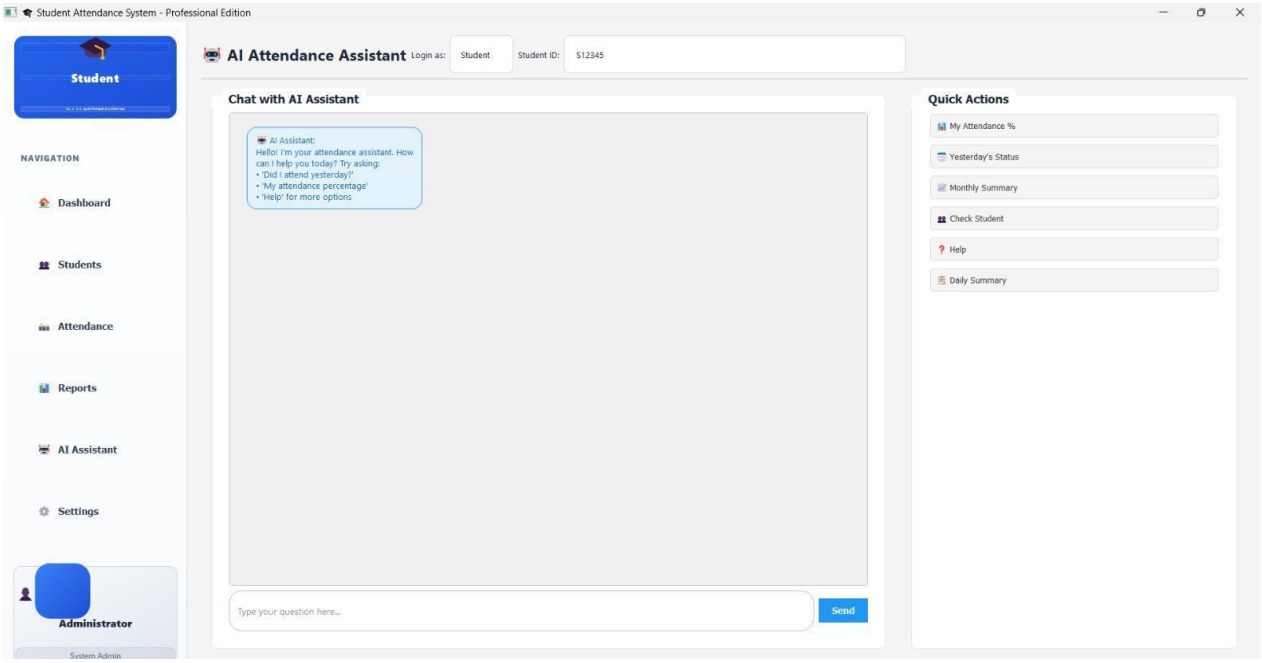
4. User Interface Module

The graphical user interface is developed using PyQt5. It provides an interactive and user-friendly environment for users to manage students, mark attendance, view reports, and interact with the system. The interface includes features such as dashboard view, student management, and report generation.



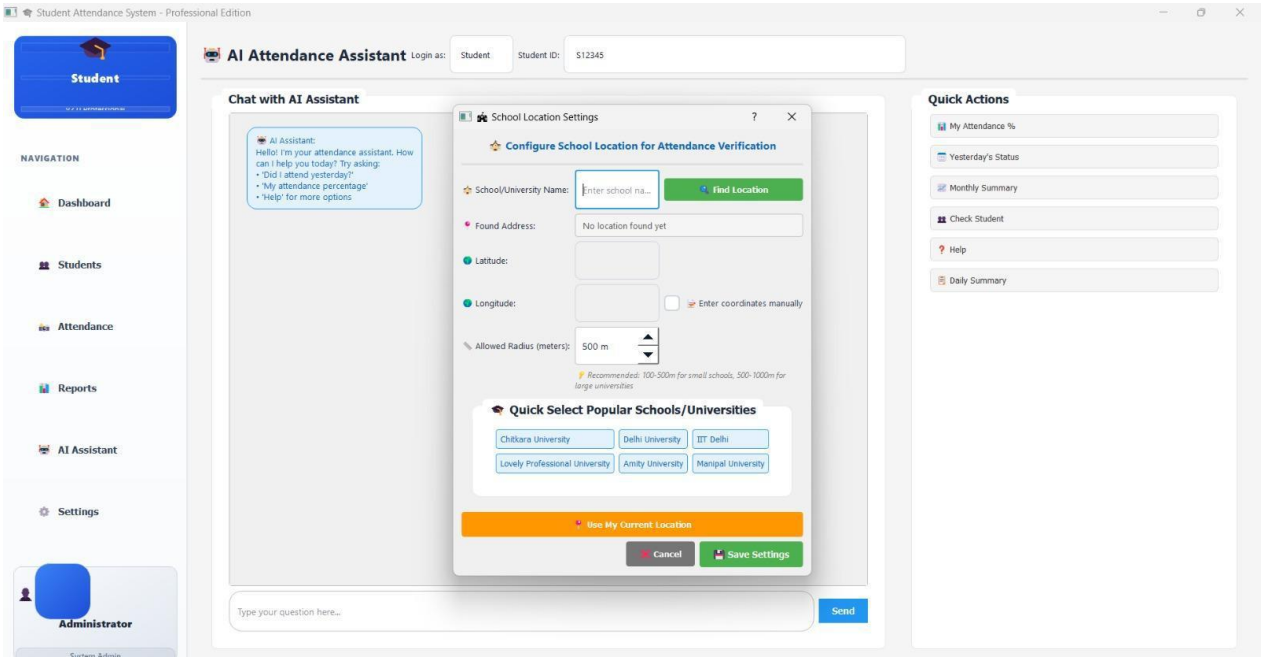
5. Chatbot Module

An AI-based chatbot is implemented to assist users in retrieving attendance-related information. The chatbot uses pattern matching and natural language processing techniques to understand user queries and provide responses such as attendance status, percentage, and absence details.



6. Location Verification Module

The system includes a location verification mechanism to ensure authenticity. It uses GPS or network-based location detection to verify that students are within the predefined classroom area. A fallback mechanism is implemented to handle cases where GPS is unavailable.



7. System Integration

All modules are integrated through the main application file, ensuring smooth communication between components. The system processes input from the camera, performs recognition and verification, stores results in the database, and displays outputs through the user interface.

8.Working Example

When a student enters the classroom, the camera captures their face. The system detects the face using OpenCV and compares it with stored student images. Once the student is recognized and verified, attendance is automatically marked with timestamp and stored in the SQLite database. The result is instantly displayed on the user interface dashboard.

MAJOR FINDINGS / RESULTS

The Smart Attend system demonstrates significant improvements over traditional manual attendance methods. The system successfully automates the entire attendance marking process, thereby reducing manual effort, minimizing human error, and saving valuable classroom time.

The facial recognition module ensures **high accuracy in student identification** by leveraging computer vision techniques, while the integration of location verification enhances system reliability by preventing proxy attendance and ensuring authenticity.

The system operates in **real-time**, enabling instant attendance marking as soon as a student is recognized. All attendance data is systematically stored in a structured database, which allows efficient retrieval, updating, and report generation.

Furthermore, the system provides easy-to-understand attendance reports and analytics, which help in monitoring student attendance trends effectively. Overall, Smart Attend improves **efficiency, accuracy, security, and transparency** in attendance management compared to traditional methods.

CONCLUSION AND FUTURE SCOPE

The Smart Attend system provides an effective and reliable solution for modern attendance management using artificial intelligence. It successfully eliminates the limitations of traditional manual attendance methods by automating the entire process and ensuring authenticity through advanced technologies such as facial recognition and location verification.

The system significantly reduces manual effort, improves accuracy, and enhances overall classroom efficiency and management. By maintaining structured digital records, it also improves transparency and makes attendance tracking more organized and efficient.

In the future, the Smart Attend system can be further enhanced to improve its performance and scalability. The accuracy of the facial recognition module can be improved under challenging conditions such as low lighting, occlusions, and varying facial angles.

Cloud-based storage can be integrated to enable real-time synchronization, better scalability, and remote access from multiple devices. Advanced analytics and visualization features can also be added to provide deeper insights into student attendance patterns and behavior trends.

The AI chatbot module can be further upgraded using more advanced natural language processing models to handle complex queries and provide a more interactive user experience. Additionally, the system can be deployed on a larger institutional scale, contributing to the development of smart campuses and automated educational infrastructure.

REFERENCES

The development of the Smart Attend system is based on various standard tools, libraries, and research materials related to artificial intelligence, computer vision, and software development. The following references were used during the design and implementation of the project:

1. OpenCV Documentation – Provides comprehensive guidance on image processing and face detection techniques. Available at: https://opencv.org
2. face_recognition Library Documentation – Used for implementing facial recognition using deep learning. Available at: https://github.com/ageitgey/face_recognition
3. Python Official Documentation – Reference for programming concepts and implementation. Available at: https://docs.python.org
4. PyQt5 Documentation – Used for developing the graphical user interface of the system.
5. SQLite Documentation – Used for database management and storage of attendance records.
6. Research papers and articles on Artificial Intelligence and Computer Vision for attendance systems.

APPENDICES

The appendices section contains supplementary information that supports the understanding and implementation of the Smart Attend system. These materials provide additional insights into the working and structure of the project.

Appendix A: System Architecture

The Smart Attend system follows a modular architecture consisting of input, processing, and output layers. The input layer includes a camera that captures real-time video. The processing layer uses facial recognition and location verification modules to identify students and validate their presence. The output layer displays attendance results and stores them in the database.

Appendix B: Database Structure

The system uses an SQLite database to store attendance records. The database contains fields such as Student ID, Name, Date, Time, and Attendance Status. This structured storage allows efficient retrieval and report generation.

Appendix C: Sample Output Screens

This section includes screenshots of the system interface, including:

- Face detection and recognition window
- Attendance marking interface
- Chatbot interaction screen
- Attendance report dashboard

Appendix D: Modules Description

The Smart Attend system consists of the following modules:

- Face Recognition Module

- Location Verification Module
- Database Management Module
- User Interface Module
- Chatbot Module

Each module works independently but is integrated to provide a seamless and efficient system.

Appendix E: Copyright Information

- Diary Number: SW-11928/2026-CO
- Status: Applied on Government Portal
- Date: 16/03/2026